

ADAPTIVE SMART GRID MANAGEMENT VIA GENTWIN & LLM INTEGRATION

A Transparent and Explainable Digital Twin Framework for Smart Grids

Thesis proposal

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Motivation — The Hidden Problem

Modern smart grids are highly optimized systems that efficiently manage electricity demand and supply.

However, from the user's perspective, these systems behave like a black box.

- Users receive electricity bills without understanding the underlying reasons
- Devices may be automatically controlled without clear explanation
- Pricing changes dynamically, but users are not informed why

This lack of transparency creates a trust gap between the system and the consumer.

Problem Statement

Current smart grid systems lack an explainability layer that connects system decisions with human understanding.

Key challenges include:

- Decision Opacity: Optimization outputs are not interpretable
- Context Overload: Massive IoT data cannot be directly understood
- No Natural Interface: Users cannot query or understand grid behavior

Research Objective

This research proposes a GenTwin-based Explainable Smart Grid Framework that enables:

- Transparent understanding of grid decisions
- Natural language interaction between user and system
- Reconstruction of “**why**” behind system actions

Unlike traditional work, this system prioritizes:

Explainability over optimization

Core Idea (Conceptual Flow)

The proposed system introduces a new pipeline:

Physical Grid → GenTwin → LLM Reasoning → Explanation → User

- Real-world data is mirrored into a digital twin
- The twin simulates and interprets system behavior
- The LLM generates human-understandable explanations

“We move from prediction to understanding.”

System Architecture Overview

The system is divided into three layers:

1. Physical Layer

- Collects real-world data from smart meters, transformers, and weather sources

2. Cyber Layer (GenTwin)

- Processes and interprets data
- Builds a digital representation of the grid

3. Application Layer

- Provides predictions, explanations, and user interaction

System Architecture Overview(contd)

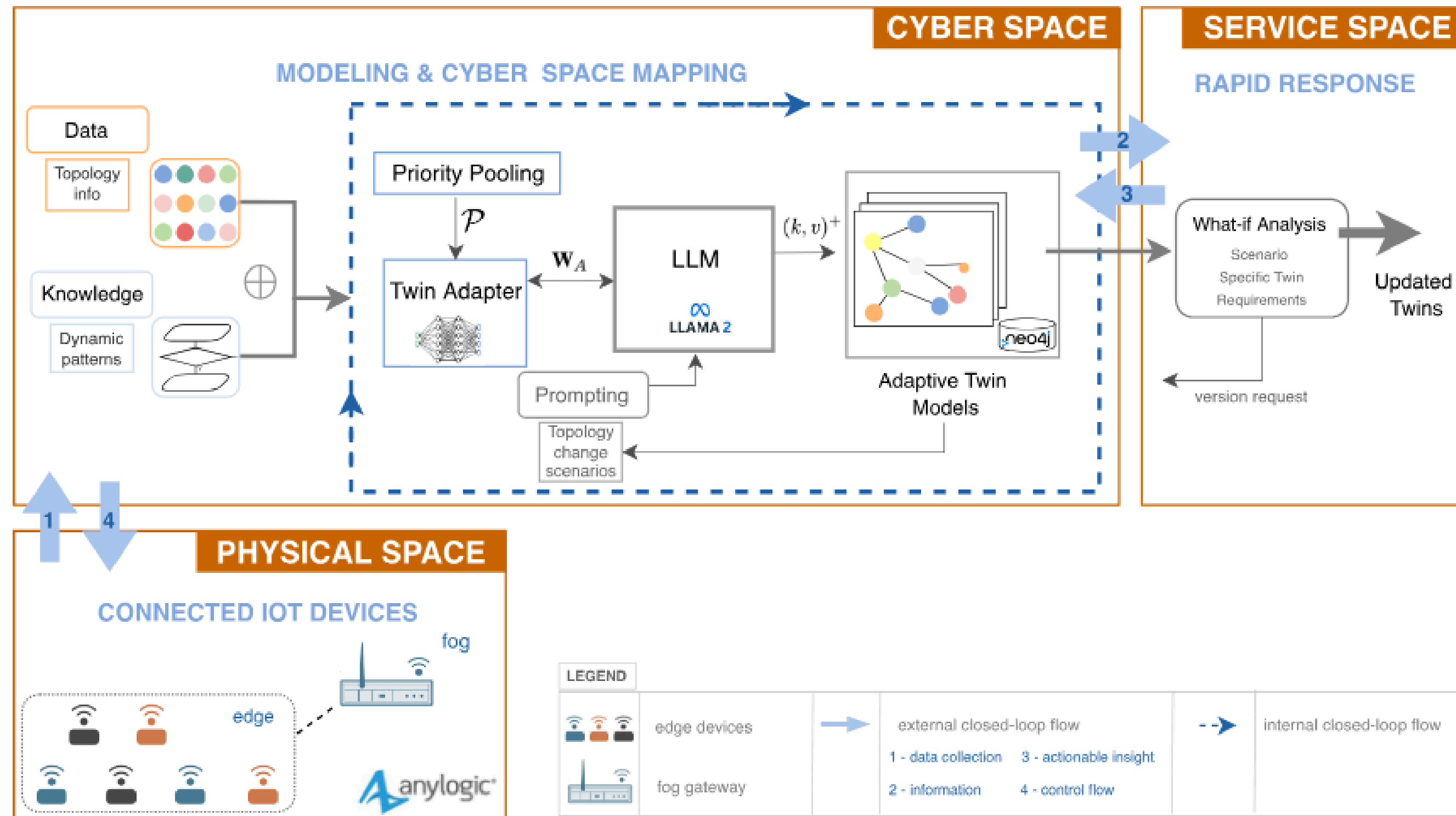
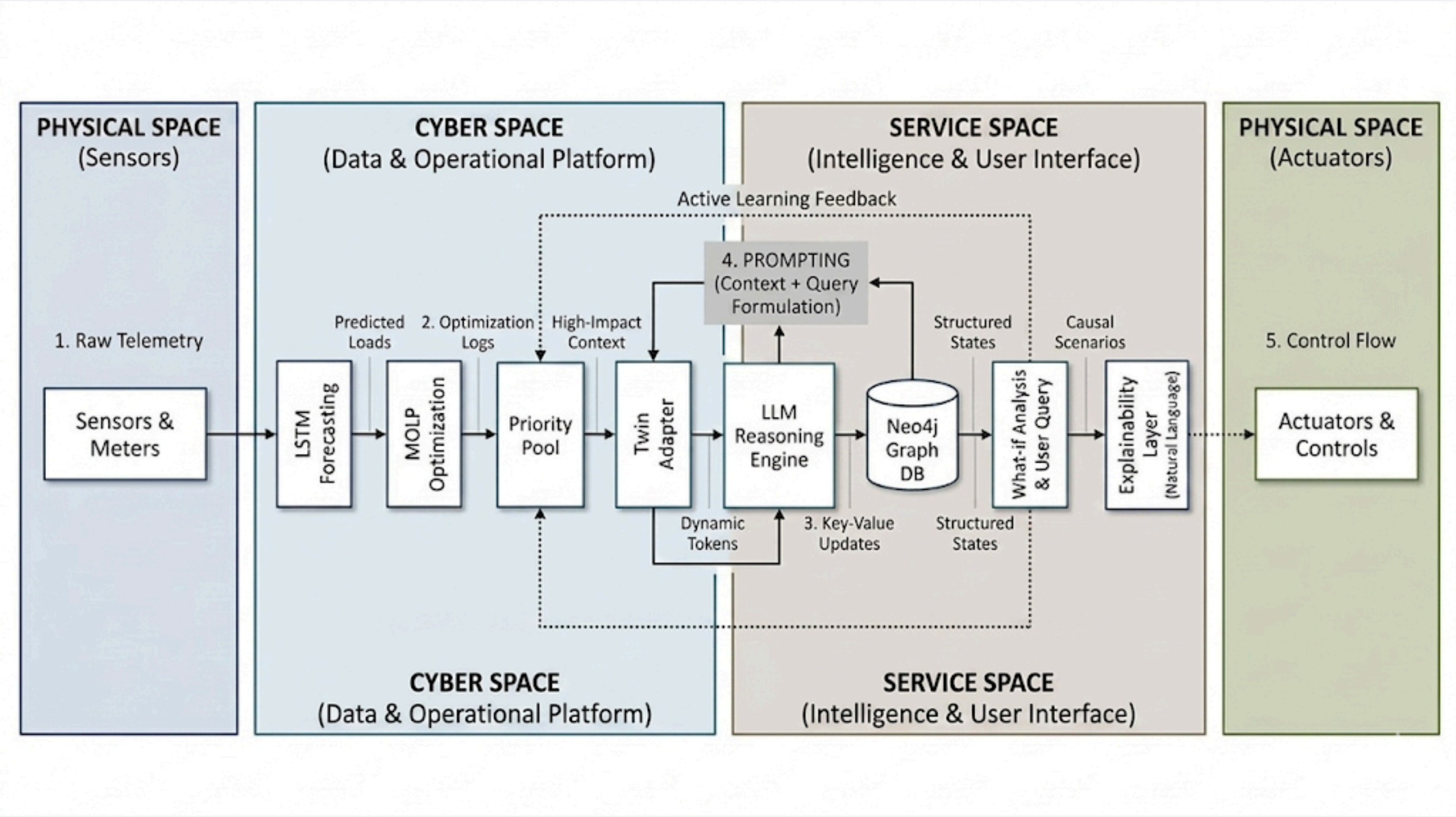


Fig. 1. Architectural framework of generative AI-aided digital twinning - GenTwin.

GenTwin Paper's proposed system architecture

Our System Architecture Overview



Our system architecture

